

Chapter 12: Sponges and Placozoans

Origin of Metazoa

- **Metazoans** are multicellular animals that fall within the **opisthokont** clade, along with fungi and **choanoflagellates**.
- **Choanoflagellates** are solitary or colonial aquatic eukaryotes with a flagellum surrounded by a collar of microvilli, strongly resembling sponge **choanocytes**.
- **Syncytial ciliate hypothesis**: Proposed that metazoans arose from a multinucleate ciliate ancestor that became compartmentalized. This hypothesis is not supported by molecular evidence or embryology.
- **Colonial flagellate hypothesis**: Proposed by Haeckel, suggesting metazoans descended from a hollow, spherical colony of flagellated cells (**blastaea**) that became differentiated. This is supported by molecular evidence placing metazoans with choanoflagellates.
- Molecular evidence suggests that **Porifera** (sponges) are the sister group to all other metazoans (**Eumetazoa**).

Phylum Porifera: Sponges

- Sponges are sessile animals with a body designed as an efficient aquatic filter.
- The name **Porifera** means “pore-bearing”, referring to the **ostia** (pores) through which water enters.
- Characteristics include:
 - Multicellular but lacking true tissues or organs.
 - Body supported by a skeleton of **spicules** and/or **spongin**.
 - Digestion is intracellular.
 - Excretion and respiration occur by diffusion.

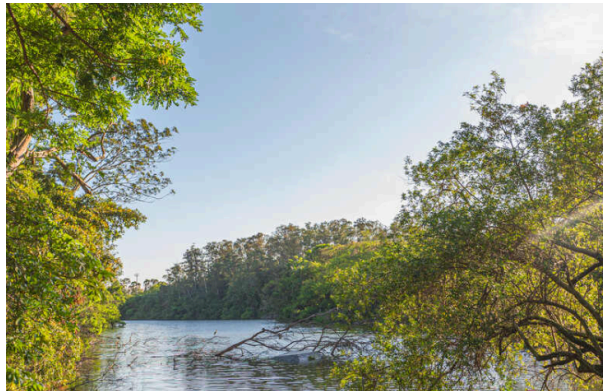


Figure 1: Some growth habits and forms of sponges. Red boring sponge, Encrusting sponge, Coral head, Finger sponge, Variable sponge, Tube sponge.

- Sponges exhibit various growth habits (encrusting, tubular, lobed) often influenced by environmental conditions like water currents.



Figure 2: Sponge choanocytes have a collar of microvilli surrounding a flagellum. Beating of the flagellum draws water through the collar (blue arrows) where food is trapped on microvilli (red arrows).

- The **choanocyte** (collar cell) is the feeding unit. The beating flagellum draws water through the collar of microvilli, which traps food particles.
- The skeletal framework consists of:
 - **Spicules**: Rigid support structures made of calcium carbonate (**calcareous**) or silica (**siliceous**).
 - **Spongin**: A form of collagen protein forming a fibrous network.

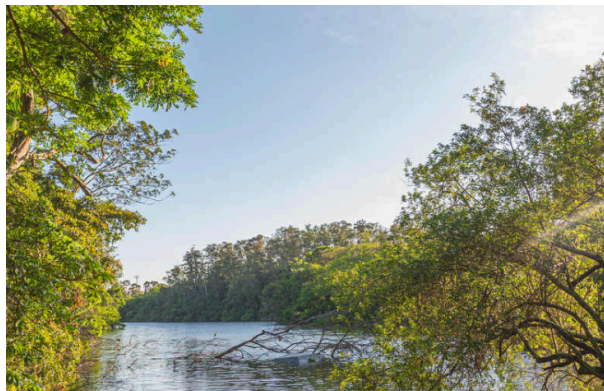


Figure 3: Diverse forms of spicules, many amazingly complex and beautiful, support a sponge body. Spongin fibers provide support in some sponges.

- Spicules come in diverse forms and are used for classification.

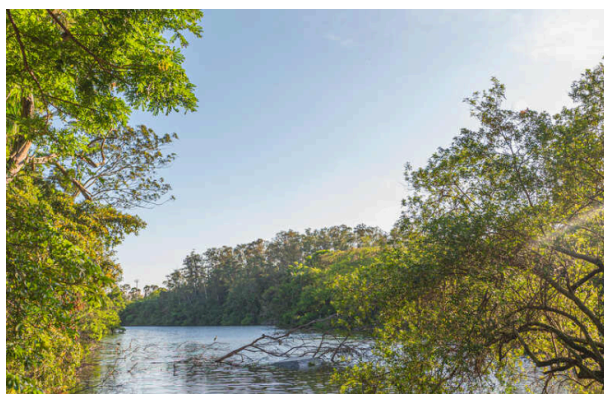


Figure 4: Cladogram depicting evolutionary relationships among the three classes of sponges with living representatives.

- Living poriferans are assigned to three classes: **Calcarea**, **Hexactinellida**, and **Demospongiae**.

Form and Function

- Sponges feed by collecting suspended particles from water pumped through internal canal systems.
- Water enters through **dermal ostia** in the outer **pinacoderm**.
- **Suspension feeding**: Nonselective consumption of particles (0.1 to 50 μm).
- **Phagocytosis**: Used for smallest particles (80% of organic carbon).
- **Pinocytosis**: Used for protein molecules.

Types of Canal Systems

- There are three main designs based on the placement of choanocytes: **Asconoid**, **Syconoid**, and **Leuconoid**.



Figure 5: Three types of sponge structure. The degree of complexity from simple asconoid to complex leuconoid type has involved mainly the water-canal and skeletal systems, accompanied by outfolding and branching of the collar-cell layer. The leuconoid type is considered the major plan for sponges, for it permits greater size and more efficient water circulation.

- **Asconoid**: Simplest organization. Choanocytes line the large central cavity (**spongocoel**). Water enters through ostia and exits through a single **osculum**. Limited in size due to surface-to-volume ratio constraints. Found only in class **Calcarea**.



Figure 6: *Clathrina canariensis* (class Calcarea) is common on Caribbean reefs in caves and under ledges.

- **Clathrina** is an example of an asconoid sponge with intertwined tubes.
- **Syconoid**: Body wall is folded to form **radial canals** lined with choanocytes. Water enters **incurrent canals**, passes through **prosopyles** into radial canals, then through **apopyles** into the spongocoel (lined with epithelial cells), and exits via the osculum.

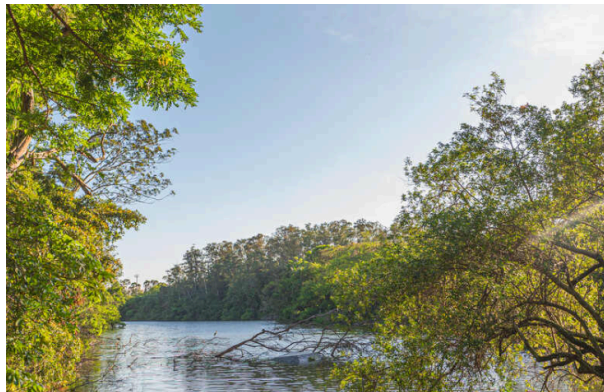


Figure 7: Cross section through wall of sponge Sycon, showing choanocytes in canals within the wall. Notice choanocytes do not line spongocoel.

- **Sycon** is a typical syconoid sponge. Note that choanocytes are restricted to the canals.
- **Leuconoid**: Most complex and common type. Choanocytes line small **flagellated chambers**. Clusters of chambers are filled from incurrent canals and discharge into excurrent canals leading to oscula. This design maximizes the surface area for feeding and allows for large body size.



Figure 8: This orange demosponge, *Mycale laevis*, often grows beneath platelike colonies of the stony coral, *Montastrea annularis*. The large oscula of the leuconoid canal system are seen at the edges of the plates. Unlike some other sponges, *Mycale* does not burrow into the coral skeleton and may actually protect coral from invasion by more destructive species. Pinkish radioles of a Christmas tree worm, *Spirobranchus giganteus* (phylum Annelida, class Polychaeta) also project from the coral colony. An unidentified reddish sponge can be seen to the right of the Christmas tree worm.

- **Leuconoid** sponges, like this **Mycale**, often have multiple oscula and can grow to large sizes.

Types of Cells in the Sponge Body

- Sponge cells are arranged in a gelatinous matrix called **mesohyl** (or mesenchyme).

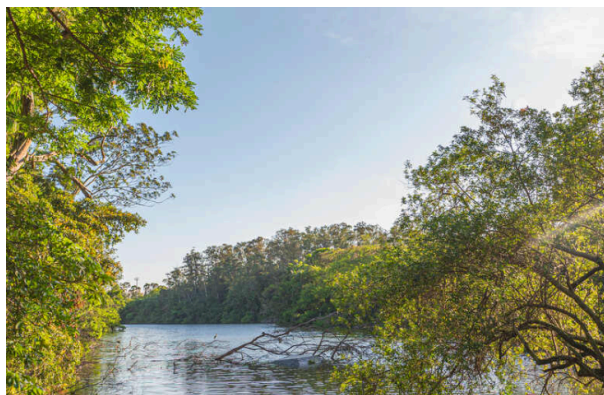


Figure 9: Small section through sponge wall, showing four types of sponge cells. Pinacocytes are protective and contractile; choanocytes create water currents and engulf food particles; archaeocytes have a variety of functions; collencytes secrete collagen.

- Major cell types include:
 - **Choanocytes**: Flagellated collar cells for water flow and feeding.
 - **Archaeocytes**: Ameboid cells in mesohyl; perform digestion (receive food from choanocytes) and can differentiate into other cell types (**sclerocytes** secrete spicules, **spongocytes** secrete spongin, **collencytes** secrete collagen).
 - **Pinacocytes**: Thin, flat, epithelial-like cells covering the exterior (**pinacoderm**) and some interior surfaces. Some are modified as contractile **myocytes**.



Figure 10: Food trapping by sponge cells. A, Cutaway section of canals showing cellular structure and direction of water flow. B, Two choanocytes and C, structure of the collar. Small red arrows indicate movement of food particles.

- Food trapping mechanism: The collar consists of microvilli connected by microfibrils. Water is pulled through this sieve, and particles are trapped and phagocytized. Digestion is entirely **intracellular**.

Cell Independence: Regeneration and Somatic Embryogenesis

- Sponges have high regenerative ability.
- **Somatic embryogenesis**: Complete reorganization of structure and function of cells from fragments or aggregates to form a new sponge.
- Asexual reproduction occurs by:
 - Fragmentation/Regeneration.
 - Budding (external).
 - **Gemmules** (internal buds).

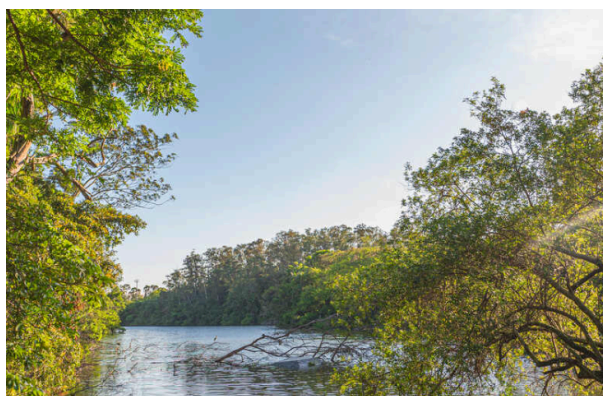


Figure 11: Section through a gemmule of a freshwater sponge (Spongillidae). Gemmules are a mechanism for survival of the harsh conditions of winter. On return of favorable conditions, the archaeocytes exit through the micropyle to form a new sponge. The archaeocytes of the gemmule give rise to all cell types of the new sponge structure.

- **Gemmules** are dormant structures formed by freshwater sponges (and some marine) to survive harsh conditions (e.g., winter, drought). Archaeocytes are enclosed in a tough spongin coat with spicules. They emerge through the **micropyle** when conditions improve.

Sexual Reproduction

- Most sponges are **monoecious** (hermaphroditic).
- Sperm arise from transformed choanocytes; oocytes from choanocytes or archaeocytes.
- Most are **viviparous** (retain zygote, release ciliated larva). Some are **oviparous**.
- Larval types:
 - **Parenchymula**: Solid-bodied, free-swimming larva (most sponges).



Figure 12: A, Development of demosponges. B, Development of the calcareous syconoid sponge Sycon.

- In **Calcarea** and some **Demospongiae**, a hollow **stomoblastula** undergoes **inversion** (turning inside out) so flagellated cells are on the outside (**amphiblastula**). Upon settlement, flagellated **micromeres** invaginate to become internal choanocytes/archaeocytes.

Class Calcarea (Calcispongiae)

- Spicules of **calcium carbonate** (CaCO_3).
- Spicules are straight (monaxons) or have 3 or 4 rays.
- Can be asconoid, syconoid, or leuconoid.
- All marine.

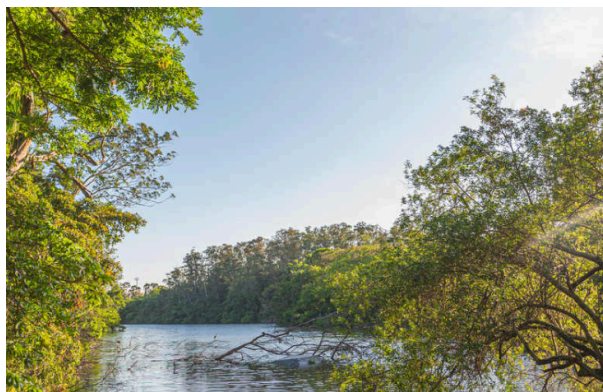


Figure 13: Some sponge body forms. Euplectella is in Hexactinellida, Poterion and Callyspongia are members of Demospongiae, and Leucosolenia is in Calcarea.

- Examples include **Leucosolenia** (asconoid), **Sycon** (syconoid), and **Clathrina**.

Class Hexactinellida (Hyalospongiae): Glass Sponges

- Deep-sea sponges.
- Skeleton of six-rayed **siliceous** spicules, often fused into a glasslike lattice.
- Body is a single continuous **syncytial** tissue called a **trabecular reticulum**.



Figure 14: Diagram of part of a flagellated chamber of hexactinellids. The primary and secondary reticula are branches of the trabecular reticulum, which is syncytial. Cell bodies of the choanoblasts and their processes are borne by the primary reticulum and are embedded in a thin, collagenous mesohyl. Processes of the choanoblasts end in collar bodies, whose collars extend up through the secondary reticulum. Flagellar action propels water (arrows) to be filtered through the mesh of collar microvilli (see Figure 12.10).

- Flagellated chambers are formed by the syncytial reticulum. **Choanoblasts** (not single cells) produce collar bodies.
- Example: **Euplectella** (Venus's flower basket).

Class Demospongiae

- Contains 95% of living sponge species.
- Skeleton of **siliceous** spicules (not 6-rayed), **spongin**, or both.
- All are **leuconoid**.
- Includes the "horny sponges" (bath sponges) which lack spicules.
- Includes the only freshwater family (**Spongillidae**).



Figure 15: Marine Demospongiae on Caribbean coral reefs. A, *Pseudoceratina crassa* is a colorful sponge growing at moderate depths. B, *Aplysina fistularis* is tall and tubular. C, *Monanchora unguifera* with commensal brittle star, *Ophiothrix suensoni* (phylum Echinodermata, class Ophiuroidea).

- Marine demosponges show great variety in color and shape.

Phylogeny and Adaptive Diversification

- **Phylogeny:** Molecular data suggests **Calcarea** may be a separate clade from **Silicea** (Demospongiae + Hexactinellida), or Porifera might be paraphyletic.
- **Adaptive Diversification:** Centered on water-current system complexity.
- **Carnivorous sponges:** A family of deep-water demosponges has lost choanocytes and canals. They use hooklike spicules to capture crustacean prey and digest them extracellularly.

Phylum Placozoa

- Contains a single established species: ***Trichoplax adhaerens***.
- Tiny (2-3 mm), marine, platelike body.
- No symmetry, no organs, no muscular or nervous system.
- Glides over food, secretes enzymes, and absorbs products.

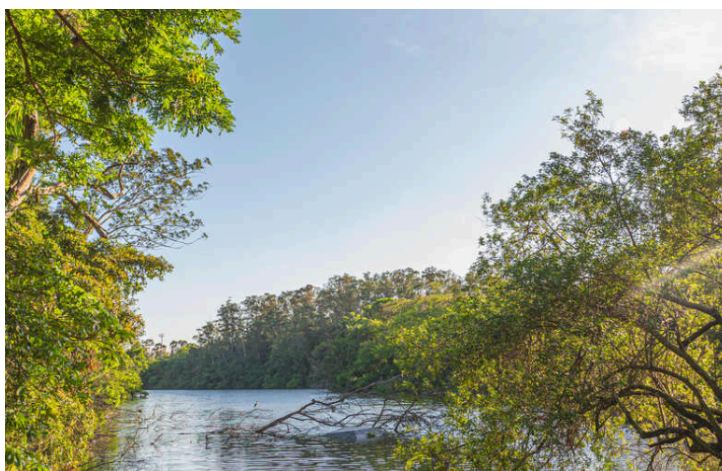


Figure 16: A, *Trichoplax adhaerens* is a marine, platelike animal only 2 to 3 mm in diameter. B, Section through *Trichoplax adhaerens*, showing histological structure.

- Structure: Dorsal epithelium (cover cells, shiny spheres) and ventral epithelium (monociliated cylinder cells, gland cells). Intermediate layer contains fibrous syncytium.